Testing across repositories

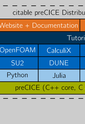
Mobile-friendly version of poster submitted to the dkRSE24 Conference 2024. March 5-7, Würzburg

Plain language summary

Single computer models are often not enough to describe phenomena and processes across science and engineering. The software library preCICE[1] allows programmers and users to combine existing models with minimal effort, producing accurate results fast. Integrations with end-user simulation codes are developed in multiple GitHub repositories, and useful simulations can only be executed by combining multiple codes. To test that development changes do not affect the numerical results, we create reproducible environments from a collection and record data for the versions of interest, connect them, and check the results at the interface between the domains, solving several challenges on the way.

A partitioned simulation example

Flow over a heated plate: A preCICE tutorial.[2] Choose your "hot solid" code among FEniCS, Nutils, CalculiX, OpenFOAM, and more. Simulate the flow with OpenFOAM or your own code.



Coupling via preCICE

Add calls to the preCICE library into each code, or use provided adapters. Start each code in a separate process (e.g., two terminals).

```
import precice
v = precice.set_mesh_vertices(positions)
precice.initialize()
while precice.is_coupling_ongoing():
    solve_as_usual()
    precice.write_data("temperature", v)
    precice.advance()
    precice.read_data("heat-flux", v)
```

An ecosystem of components

citable preCICE Distribution (on DaRUS [3])

Website + Documentation	Nix pkgs	VM
Tutorials		
OpenFOAM	CalculiX	FEniCS	deal.II
SU2	DUNE	DuMuX	MBDyn
Python	Julia	Matlab	*Rust*
preCICE (C++ core, C & Fortran bindings)		

adapters for

bindings for core

Reference results and comparisons

Stored next to the cases

Git Large File Storage

git-lfs.com

```
flow-over-heated-plate/
├── solid-fenics/
├── solid-nutils/
├── reference-results/
│   ├── fluid-openfoam_solid-fenics.tar.gz
│   └── fluid-openfoam_solid-nutils.tar.gz
```

VTK compared with

FieldCompare

gitlab.com/dglaser/fieldcompare

```
# One directory for each example
# One directory for each code

# Multiple code options
# Reference results versioned
# One file for each combination
```

fluid-openfoam	preCICE	solid-fenics
OpenFOAM output: foam_logs	mesh exports (.vtk) iterations.log	FEniCS output: vtk, logs

specific to OpenFOAM, controlled by others

standard formats (VTK, CSV)

specific to FEniCS, controlled by others

Metadata, tests, and test suites

metadata.yaml (precice/tutorials/flow-over-heated-plate/)

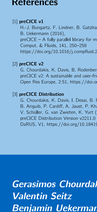
```
name: flow over heated plate
path: flow-over-heated-plate

participants:
  - Fluid
  - Solid

cases:
  - fluid-openfoam:
      participant: Fluid
      directory: ./fluid-openfoam
      run: ./run.sh
      component: openfoam-adapter
  - solid-fenics:
      participant: Solid
      directory: ./solid-fenics
      run: ./run.sh
      component: fenics-adapter
```

tests.yaml (precice/tutorials)

```
test_suites:
  openfoam_adapter_gr:
    tutorials:
      - path: flow-over-heated-plate
        case_combination:
          - fluid-openfoam
          - solid-fenics
        reference_results:
          - fluid-of_solid-fenics.tar.gz
```

Multiple repos, multiple perspectives

Partitioned approach makes CI complex. Tests can run locally or on GitHub Actions, triggered manually or by pull requests.



Can we release preCICE v3?

Continue merge the PR?

openfoam-adapter-workflow.yml (precice/openfoam-adapter)

```
jobs:
  testing-approved-or:
    if: github.event.review.state == 'APPROVED'
    uses: precice/tutorials/.github/workflows/workflow.yml@develop
    with:
      suites: openfoam_adapter_gr
      build_args: OPENFOAM_ADAPTER_REF_${{github.event.pr.head.sha}}
      systests_branch: develop
```

workflow.yml (precice/tutorials)

```
jobs:
  run_testsuite:
    runs-on: [self-hosted, linux, x64, precice-tests-vm]
    steps:
      - name: Run tests
        run: |
          cd tools/tests
          python systemtests.py
            --build_args=${{inputs.build_args}}
            --suites=${{inputs.suites}}
```

References

[1] **preCICE v3**
H.-J. Bungartz, F. Lindner, B. Gatzhammer, M. Mehl, K. Scheufele, A. Shukaev, B. Uekermann (2016).
preCICE – A fully parallel library for multi-physics surface coupling
Comput. & Fluids, 141, 250-258.
https://doi.org/10.1016/j.compfluid.2016.04.003

[2] **preCICE v3**
H. Chourdakis, K. Davis, B. Rodenberg, M. Schulte, F. Simonis, B. Uekermann, G. Abrams, H.-J. Bungartz, L. Cheung Yau, I. Desai, K. Eder, R. Hertrich, F. Lindner, A. Rusch, D. Sashko, D. Schneider, A. Totounferoush, D. Volland, P. Vollmer, O. Weeger (2022).
preCICE v2: A sustainable and user-friendly coupling library [version 2; peer review: 2 approved]
Open Res Europe 2:51.
https://doi.org/10.12688/openreseurope.14445.2

[3] **preCICE Distribution**
H. Chourdakis, K. Davis, B. Rodenberg, M. Schulte, F. Simonis, B. Uekermann, G. Abrams, H.-J. Bungartz, L. Cheung Yau, I. Desai, K. Eder, R. Hertrich, F. Lindner, A. Rusch, D. Sashko, D. Schneider, A. Totounferoush, D. Volland, P. Vollmer, O. Weeger (2022).
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System regression tests for the preCICE partitioned simulation ecosystem

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